Program : **Diploma in Chemical Engineering**	
Course Code : **6077**	Course Title: **Mass Transfer Lab**
Semester : **6**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- To enable the students in interpreting the concepts of mass transfer
- To provide hands-on experience in mass transfer operations

Course Pre-requisites:

Topic	Course code	Course name	Semester
Mass transfer operations – diffusion, drying		Mass Transfer Operations I	4
Mass transfer operations – diffusion, distillation, leaching, extraction		Mass Transfer Operations II	5

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the principles of distillation.	12	Applying
CO2	Interpret the principles of leaching	12	Applying
CO3	Interpret the principles of liquid – liquid extraction.	6	Applying
CO4	Interpret the principles of humidification and drying	15	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Interpret the principles of distillation**		
M1.01	Plot vapour liquid equilibrium curve for simple distillation of ethanol-water system of given composition.	3	Applying
M1.02	Verify Rayleigh's equation and material balance for simple distillation.	3	Applying
M1.03	Estimate the vaporization efficiency for steam distillation	3	Applying
M1.04	Open ended experiment	3	Applying
CO2	**Interpret the principles of leaching**		
M2.01	Estimate the percentage recovery for single stage leaching.	3	Applying
M2.02	Estimate the percentage recovery for multi stage leaching.	3	Applying
M2.03	Determine percentage recovery of oil from oilseed by cross current leaching.	3	Applying
M2.04	Open ended experiment	3	Applying
	Series Test – I		
CO3	**Interpret the principles of liquid – liquid extraction**		
M3.01	Plot binodal solubility curve for the given ternary liquid system	3	Applying
M3.02	Open ended experiment	3	Applying

CO4	**Interpret the principles of humidification and drying**		
M 4.01	Estimate the humidity of air using dew point method.	3	Applying
M 4.02	Estimate the humidity of air using wet and dry bulb thermometers	3	Applying
M 4.03	Estimate the rate of drying for the given sample using batch dryer	3	Applying
M 4.04	Examine the effect of inlet water temperature in cooling tower performance.	3	Applying
M 4.05	Examine the effect of air flow rate in cooling tower performance.	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Unit Operations. II; K A Gavhane
R1	Mass Transfer Operations; Treybal.R.E
R2	Unit Operations; McCabe and Smith
R3	Mass Transfer Operations; Surya Narayana

Online Resources

Sl.No	Website Link
1	https://www.vlab.co.in/broad-area-chemical-engineering